Cart-Pole Inverted Pendulum: Simulation and PID Control

Project Overview

This project focuses on the mathematical modelling, dynamic simulation and PID-based control of a cart-pole inverted pendulum system.

The system consists of a pendulum mounted on a horizontally moving cart. The main objective is to maintain the pendulum in its unstable upright position while controlling the horizontal movement of the cart.

The project was developed using MATLAB Simulink for system modelling and control simulation, with SolidWorks used for mechanical CAD design.

Project Objectives

- Develop a dynamic model of the cart-pole inverted pendulum.
- Analyse the nonlinear behaviour of the system.
- Design separate PID controllers for cart position and pendulum angle.
- Implement the control architecture in MATLAB Simulink.
- Evaluate system performance through simulation.
- Develop a 3D mechanical design using SolidWorks.

Software and Tools

- MATLAB Simulink
- SolidWorks
- PID Control
- Dynamic System Modelling
- Simulation and Response Analysis

Project Components

1. Simulink Model

Dynamic system modelling and implementation of the PID control architecture.

2. Mechanical CAD Design

3D modelling and mechanical assembly using SolidWorks.

3. Simulation and Results

Analysis of pendulum angular response, cart position response and system movement.

Simulink Modelling and PID Controller Design

MATLAB Simulink was used to implement the mathematical model and develop the control system.

The system model represents the relationship between the applied control force, cart movement and pendulum angular response.

Two PID controllers were used in the control architecture:

Pendulum Angle PID Controller

The first PID controller is responsible for regulating the pendulum angle. Its main objective is to maintain the pendulum close to its upright reference position.

The controller compares the reference angle with the measured or simulated pendulum angle. The resulting error is processed by the proportional, integral and derivative components to generate a control action.

Cart Position PID Controller

The second PID controller is responsible for regulating the horizontal position of the cart.

It compares the desired cart position with the actual cart position and generates a control correction based on the position error.

The two control actions are coordinated through the implemented Simulink control architecture to achieve pendulum stabilisation and cart position regulation.

The PID control law:

$$u(t) = Kp\, e(t) + Ki \int e(t)dt + Kd\, de(t)/dt$$

Where:

- Kp is the proportional gain.
- Ki is the integral gain.
- Kd is the derivative gain.
- e(t) is the control error.
- u(t) is the controller output.

The controller gains influence system stability, response speed, overshoot and steady-state error. Appropriate tuning is therefore important for achieving satisfactory control performance.